

OMKAR CHAITHANYA R

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Location: Bengaluru

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TECHNICAL SKILLS

- **AI | ML | DL:** PyTorch, Machine Learning, Deep Learning (CNN), NumPy, Pandas, Matplotlib
- **Python Libraries:** Fast API, Flask, OpenCV, Pandas, Poetry, Lang chain, Llama Index
- **Web Frameworks:** React.js, Next.js, Express, Node.js, Vue.js, Bootstrap, Tailwind
- **Databases:** MongoDB, PostgreSQL, Firebase, Redis, MySQL
- **Languages:** HTML, CSS, JavaScript, TypeScript, Python, C/C++, SQL

PUBLICATIONS

- HERITAGE SITE IMAGE RESTORATION USING AI IMAGE GENERATION 2025
- REAL TIME ANIMAL INTRUSION MONITORING (ONGOING)

PROJECTS

REAL-TIME ANIMAL INTRUSION MONITORING (ongoing)

- Real-Time AI Detection Pipeline: Developed a wildlife monitoring system using YOLOv8 to detect species (elephants, cattle) with high accuracy, utilizing Python and deep learning to automate intrusion alerts in rural areas.
- IoT & Low-Power Engineering: Optimized system efficiency by integrating PIR/Vibration sensors for triggered image capture and implemented LoRa/GSM protocols to ensure reliable, long-range data transmission in low-connectivity zones.

HERITAGE SITE IMAGE RESTORATION USING GENERATIVE AI

- Researched and showcased the cultural and historical significance of the heritage site to highlight its legacy.
- Applied Stable Diffusion with Realistic Vision and Soft Edge detection to generate and enhance visuals of the site's architecture and traditions

DISEASE DETECTION IN PLANT LEAVES USING CNN

- Built a plant disease detection model using Python, TensorFlow, and Keras to classify leaf diseases.
- Trained a CNN model on a large dataset to accurately predict plant diseases and support effective crop health monitoring.

DROWSINESS DETECTION USING IOT

- Developed a driver drowsiness detection system using Arduino Mega, OpenCV, and deep learning.
- Implemented facial landmark detection to identify early signs of fatigue, such as eye closure and head movements, ensuring accurate monitoring.

AMD MI300X x8 GPU Optimization

- Optimization of distributed dense matrix multiplication algorithms for optimal usage in modern GPU clusters using Pytorch & Triton on the cloud GPU.

EXTRA CURRICULAR

Hackathons

Participant

- Participated in LMA EDGE Hackathon
- Participated in Arm AI Developer Challenge
- AMD AI Sprint Hackathon 2025 edition

EDUCATION

- BTech, Computer Science, Dayananda Sagar University
- 11th/12th, Physics | Chemistry | Mathematics | Computer Science, Vijaya Composite PU College
- Schooling, CBSE Syllabus, jnanodaya school

CERTIFICATIONS

- HP Life Data Science & Analytics – Sep'25 ([link](#))
- Oracle Cloud Infrastructure – Oct'08 ([link](#))
- Linux Programming – Jan'2 ([link](#))

